

Element F: Consideration of Design Viability

The development office at our independent boarding and day school has an overflow of items. The excess items piled on the floor hinder workflow and efficiency. Many items move around the development office; therefore, it is important to keep them organized. The current storage situation is unsatisfactory in the eyes of school donors and visitors.

Our proposed solution has shelves that are taller and deeper than their existing boxes, so they will fit on the shelf. There is a space for a student intern to work, and space for a coffee station. The doors will hide items from view and it will keep the office visually appealing.

We have the needed tools to build our solution in our engineering lab and woodshop. Students on our team have taken our woodshop class and so have learned safe use of machines and hand tools, and construction techniques.

To go beyond our stakeholder's requirements, we also included an intern desk and concealed a non-functional door. The extra space in the intern desk is used to have more space for work and makes the intern more comfortable. Covering the door makes the office look nicer.

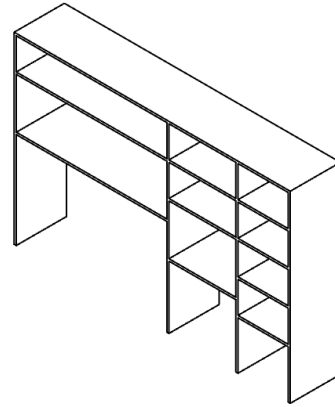
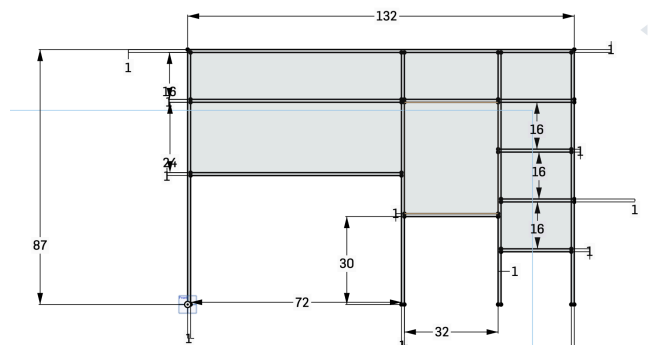
The Director of the Performing Arts at our independent private boarding and day school, Mr. C., who has substantial construction experience, and frequently helps with set design for drama productions, confirmed our use of structural principles and offered input on the practicality of our material choices.

Element G: Construction of a Testable Prototype

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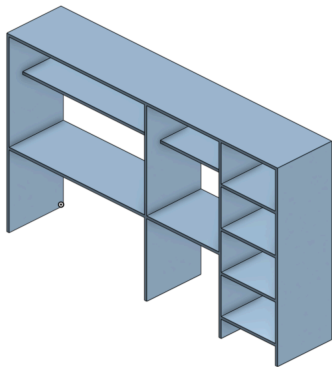
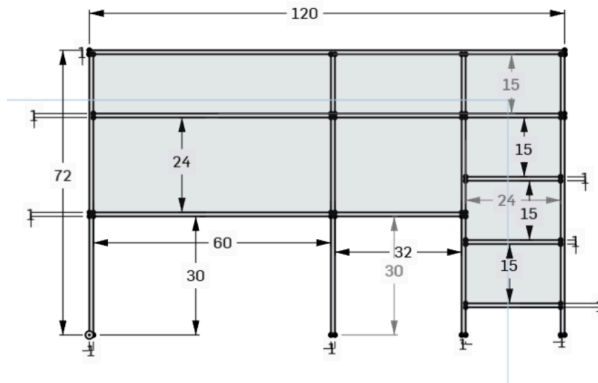
Our top three design priorities focused on creating concealed storage for better organization, ensuring the desk could hold a variety of items, and maintaining easy access to outlets and windows to support everyday functionality. To develop our design efficiently and avoid wasting materials, we used OnShape to model both our initial, first, and final prototypes. This digital platform allowed us to visualize and refine our ideas early in the process, which helped reduce the risk of failure during physical construction.

We built our first prototype using OnShape after conducting group discussions about the desk's purpose and how it could be safely and structurally designed. These conversations included our team and the stakeholders who would be using the final product. Our initial sketches and ideas evolved significantly through this process. For example, we increased the height of the far-left shelving unit to accommodate more storage boxes and extended the desk's length from 120 inches to 132 inches. We also raised the overall height from 72 inches to 87 inches, allowing us to conceal a nonfunctional door as requested by the stakeholders. These changes made the desk more practical, spacious, and better suited to the room's layout.



First OnShape Design (First Prototype)

After finalizing the measurements and adjusting the design, we created a second, more refined prototype in OnShape, which became the foundation for our final build. We chose wood as the primary construction material, based on feedback from the office staff. The modeling process allowed us to present a clear three-dimensional representation of the cabinet, which helped us gather suggestions for improvement. Based on this feedback, we redesigned the cabinet to better meet the needs of those who would be using it.

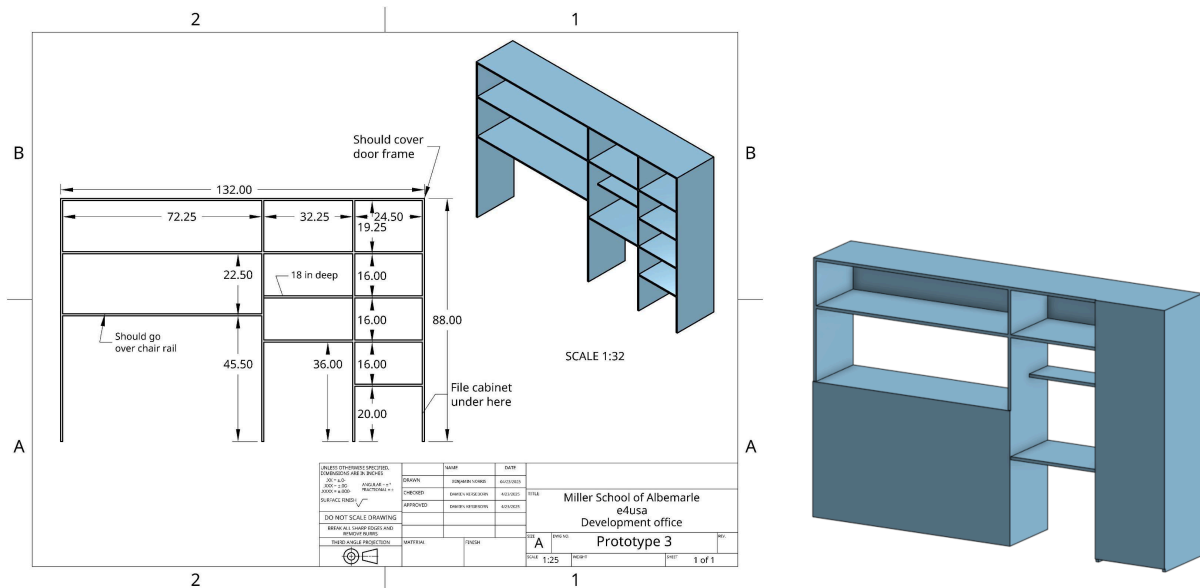


Second OnShape Design (Second Prototype)

Throughout the process, we worked closely with four experts: Mr. C, Mr. H, Mr. F, and Mr. M. They provided valuable insights based on their backgrounds in engineering, woodworking, and construction. Their feedback was especially helpful when developing our testing approach. They recommended adding support boards behind the desk to ensure it could support heavy materials while keeping the outlets accessible.

Our final prototype, now measuring 132 inches in length and 87 inches in height, includes enough space to fit a file cabinet below the student work area if needed. We plan to paint the desk white to ensure it matches the room's overall aesthetic. This final design not only reflects the collective input of our team, the practical needs of the space, and the priorities of

those who will use it, but also ties back to our decision matrix. The stakeholders requested doors to cover the shelves and provide storage.



Final OnShape Design (Third Prototype)

We did not build a physical prototype due to time constraints, but our maintenance team will build it over the summer. To build our design, we would begin by measuring and cutting three vertical pieces of sanded plywood (88"x24") to form the frame. Next, we would cut the top piece of wood (132"x24") and prepare two large shelves (72.25"x24"), as well as 3 medium shelves (32.25"x24"). One medium shelf would be cut to a smaller depth (32.25"x18"). We would also cut four small shelves (24.5"x24") to complete the structure. After that, the top piece would be attached to the vertical pieces using brackets, and the shelves would be slid into place and secured. We would then measure and cut melamine for the back of the top shelves. (132"x20"). The final steps would involve applying veneer to the edges, sanding rough areas, and painting the entire shelf to match the existing wall paint color.

Happy Hoppy Frogs
Development Office
Spring 2025

We will be able to test for accessibility of outlets, if boxes can be concealed, amount of storage space, and if a coffee station will fit. Because we are not building a physical prototype, we will not be able to test for if it is comfortable and inviting to guests, the aesthetics, display, We will not test for lockable storage because we will not be building a solution for the hallway.